Source-Based AI Recommendation Explanations — Experiment Spec v1

1. Working Title

Source-Based AI Recommendation Explanations and Consumer Decision-Making on E-Commerce Platforms

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2. Research Question

How does the information source disclosed in an AI-generated recommendation explanation influence consumers' perceptions of relevance, source credibility, privacy concern, trust, and willingness to rely on the recommendation?

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3. Core Research Idea

AI recommendation systems can explain not only *why* a product is recommended, but also *what information source* was used to generate that recommendation.

Different information sources may produce different psychological responses.

For example:

- behavioral information may make a recommendation feel more personally relevant but also increase privacy concern;
- customer-review information may increase perceived source credibility;
- product-feature information may increase perceived fit with stated preferences;
- combining multiple sources may amplify benefits or introduce additional complexity and privacy concerns.

The study therefore examines **source disclosure as a design feature of AI recommendation explanations**.

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4. Experimental Design

Design

Five-condition between-subjects online experiment.

Participants will be randomly assigned to one of five explanation conditions.

Product Context

Participants imagine shopping online for wireless headphones.

The AI shopping assistant recommends the same fictional product in every condition:

AeroSound X5 Wireless Headphones

The product, recommendation, product attributes, tone, and overall presentation remain constant across conditions.

The primary experimental manipulation is the **information source disclosed in the recommendation explanation**.

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5. Experimental Conditions

Condition 1 — Control

The AI recommends the product without identifying a specific information source underlying the recommendation.

Condition 2 — Behavior-Based

The AI explains that the recommendation is based on the participant's recent browsing and shopping activity.

Condition 3 — Review-Based

The AI explains that the recommendation is based on patterns found in other consumers' reviews.

Condition 4 — Product-Feature-Based

The AI explains that the recommendation is based on a match between the product's attributes and the preferences stated by the participant.

Condition 5 — Mixed-Source

The AI explains that the recommendation combines behavioral information, customer-review information, and product-feature information.

Stimulus Reference

See separate document: ***Source-Based – Stimuli v2***

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6. Independent Variable

Recommendation Explanation Source

Five levels:

1. Control
2. Behavior-Based
3. Review-Based
4. Product-Feature-Based
5. Mixed-Source

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7. Core Psychological Mechanisms

Perceived Relevance

The extent to which the recommendation appears to fit the participant's needs and preferences.

Perceived Source Credibility

The extent to which the information underlying the recommendation is perceived as credible and reliable.

Situational Privacy Concern

Concern or discomfort triggered by the collection or use of personal or behavioral information in generating the recommendation.

Trust

Confidence in the AI shopping assistant and its recommendation.

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8. Outcome Variables

Primary Psychological Outcome

Trust in the AI Recommendation

Reliance Outcomes

Reliance Intention
Willingness to use or follow the AI recommendation.

Behavioral Reliance
Whether the participant chooses the AI-recommended product or continues searching.

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9. Conceptual Model

Behavior-Based Explanation
→ Perceived Relevance (+)
→ Trust
→ Reliance

Behavior-Based Explanation
→ Privacy Concern (+)
→ Trust (−)
→ Reliance

Review-Based Explanation
→ Source Credibility (+)
→ Trust
→ Reliance

Product-Feature-Based Explanation
→ Perceived Relevance (+)
→ Trust
→ Reliance

Mixed-Source Explanation
→ Exploratory effects on relevance, credibility, privacy concern, trust, and reliance

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10. Hypotheses

H1 — Perceived Relevance

- **H1a.** A behavior-based recommendation explanation will lead to higher perceived relevance than a no-source control explanation.
- **H1b.** A product-feature-based recommendation explanation will lead to higher perceived relevance than a no-source control explanation.

H2 — Privacy Concern

- **H2.** A behavior-based recommendation explanation will produce greater privacy concern than review-based, product-feature-based, and no-source explanations.

H3 — Source Credibility

- **H3.** A review-based recommendation explanation will produce greater perceived source credibility than a no-source explanation.

H4 — Psychological Mechanisms of Trust

- **H4a.** Perceived relevance will be positively associated with trust in the AI recommendation.

- **H4b.** Perceived source credibility will be positively associated with trust in the AI recommendation.
- **H4c.** Privacy concern will be negatively associated with trust in the AI recommendation.

H5 — Trust and Reliance

- **H5.** Greater trust in the AI recommendation will be associated with greater willingness to rely on the recommendation.

H6 — Indirect Effects

- **H6.** Recommendation explanation source will influence reliance indirectly through perceived relevance, source credibility, privacy concern, and subsequent trust.

RQ1 — Mixed-Source Explanation

How does a mixed-source explanation affect perceived relevance, source credibility, privacy concern, trust, and reliance relative to single-source explanations?

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11. Measurement

Perceived Relevance

3 items

Perceived Source Credibility

3 items

Situational Privacy Concern

3 items

Trust

4 items

Reliance Intention

3 items

Behavioral Reliance

1 binary decision

Manipulation Checks

5 items

See separate documents:

- **Source-Based – Measurement v1**
- **Source-Based – Manipulation Checks v1**

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12. Planned Experiment Flow

1. Informed consent
2. Shopping scenario
3. Random assignment to one of five conditions
4. AI recommendation stimulus
5. Perceived relevance
6. Perceived source credibility
7. Situational privacy concern
8. Trust
9. Reliance intention
10. Behavioral reliance decision
11. Manipulation checks
12. Demographic/control variables
13. Debrief

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13. Pilot Objectives

The pilot study will evaluate:

- whether participants correctly distinguish the five explanation-source conditions;
- whether the wording and length of the stimuli are sufficiently matched;
- whether the manipulation affects the intended perceptions rather than unintended features of the interface;
- whether multi-item measures demonstrate acceptable internal consistency;
- whether perceived relevance, source credibility, privacy concern, and trust are empirically distinguishable constructs;
- whether the survey length and instructions are manageable for participants.

No formal substantive conclusions will be drawn from the pilot.

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14. Current Status

Completed

- Conceptual Model v1

- Five-condition experimental design
- Stimuli v2
- Manipulation Checks v1
- Hypotheses v1
- Measurement v1
- Experiment Spec v1
- Analysis Plan v1

Next Step

Develop the complete Qualtrics experiment flow and prepare the pilot version.